

[illegible]

1. NSAID (angry pill)

WHAT YOU SEE

Angry orange NSAID pill spewing

WHAT IT ENCODES

NSAIDs are a leading cause of drug-induced acute interstitial nephritis (AIN). Unlike antibiotic-induced AIN, NSAID AIN tends to have a LONGER latency (months), MINIMAL allergic features (often no fever/rash/eosinophilia), and uniquely can coexist with NEPHROTIC-range proteinuria from concurrent minimal change disease (MCD). Mechanism is T-cell mediated (type IV) hypersensitivity to the drug-modified tubular antigen, not dose-dependent toxicity. NSAIDs also cause hemodynamic AKI (afferent vasoconstriction) — a separate mechanism.

BOARD TRAP

Heavy (nephrotic-range) proteinuria does not exclude AIN — NSAID AIN can present with MCD-pattern nephrotic syndrome; calling it 'primary glomerular disease' and missing the NSAID is the trap.

2. Antibiotics list

WHAT YOU SEE

A hanging scroll under a lantern listing the antibiotic culprits — beta-lactams, sulfonamides, quinolones, vancomycin, rifampin — the classic, most-common drug triggers of AIN.

WHAT IT ENCODES

Antibiotics are the classic and most common drug cause of AIN. High-yield offenders: beta-lactams (methicillin is the historic prototype; penicillins, cephalosporins), sulfonamides (including TMP-SMX), fluoroquinolones, vancomycin, and rifampin. The reaction is idiosyncratic/hypersensitivity (NOT dose-related) and can recur faster and more severely on re-exposure.

BOARD TRAP

AIN is a hypersensitivity reaction, not dose-dependent toxicity — assuming a 'normal/therapeutic dose' rules out drug AIN is wrong; even a single standard course can trigger it.

3. PD-1 checkpoint inhibitor

WHAT YOU SEE

Robot labeled PD-1

WHAT IT ENCODES

Immune checkpoint inhibitors (anti-PD-1 nivolumab/pembrolizumab, anti-PD-L1, anti-CTLA-4 ipilimumab) are an increasingly common cause of AIN as an immune-related adverse event. Onset is often DELAYED (weeks to months after starting), and concurrent PPI/NSAID use raises risk. Management may require holding the checkpoint inhibitor and high-dose glucocorticoids, often without permanent discontinuation depending on severity.

BOARD TRAP

In an oncology patient on immunotherapy, rising creatinine is easily blamed on contrast or pre-renal causes — missing checkpoint-inhibitor AIN (which needs steroids, not just fluids) is the trap.

4. PPI sleeping pharmacist

WHAT YOU SEE

Sleeping old pharmacist with mortar/pestle

WHAT IT ENCODES

Proton pump inhibitors (omeprazole and the other '-prazoles') are a very common, often OVERLOOKED cause of AIN. Onset is typically insidious/late (weeks to many months) with few classic allergic features, so it is easily missed. PPIs are also linked to chronic interstitial nephritis and CKD with prolonged use. Always review the med list for PPIs in unexplained AKI.

BOARD TRAP

Because PPIs are OTC and seem benign, they are forgotten on the med reconciliation — overlooking a chronic PPI as the AIN trigger is a classic miss.

5. Sulfonamides/cipro/rifampin

WHAT YOU SEE

List by armored knight: sulfonamides, fluoroquinolones, rifampin

WHAT IT ENCODES

Sulfonamides, fluoroquinolones, and rifampin are reinforced offenders. Note sulfonamides have a DUAL renal toxicity — hypersensitivity AIN AND crystal-induced (crystalluria/tubular obstruction) injury. Rifampin is notable for causing severe AIN especially with intermittent or interrupted-then-resumed dosing (re-challenge).

BOARD TRAP

Sulfa-related AKI can be crystal nephropathy rather than AIN — distinguish by urinalysis (crystals/hematuria with obstruction vs sterile pyuria/WBC casts of AIN); lumping them together is the trap.

6. Fever + rash + eosinophilia triad

WHAT YOU SEE

The flushed patient covered in a maculopapular rash, tagged 'MINORITY ~10%' — the classic fever+rash+eosinophilia triad, drawn small because it appears in only a minority of cases.

WHAT IT ENCODES

The classic hypersensitivity triad of AIN is FEVER + RASH (maculopapular) + EOSINOPHILIA, often with arthralgias and peripheral eosinophilia on CBC. Critically, the COMPLETE triad is present in only a MINORITY (~10% or fewer) of patients — most have one or none. Fever is the single most common component; the full triad is the exception, not the rule.

BOARD TRAP

Requiring the full triad before diagnosing AIN is wrong — the complete fever+rash+eosinophilia combination occurs in only ~10% of cases, so its absence cannot be used to exclude AIN.

7. NO RASH NO EOS

WHAT YOU SEE

Red crossed-out sign 'NO RASH NO EOS'

WHAT IT ENCODES

Absence of rash and a normal eosinophil count do NOT rule out AIN — these features are insensitive, and many biopsy-proven cases (especially NSAID- and PPI-induced) lack them entirely. The diagnosis rests on the clinical picture (offending drug + AKI + bland-to-pyuric sediment) and is confirmed by renal BIOPSY when needed. Maintain a low threshold to suspect AIN in unexplained AKI.

BOARD TRAP

'Eosinophils are normal, so it can't be AIN' is a classic distractor — normal eosinophils (and no rash) absolutely do not exclude AIN; biopsy is the gold standard.

8. Urine eos not useful

WHAT YOU SEE

Flask 'URINE EOS NOT USEFUL', WBC casts

WHAT IT ENCODES

Urine eosinophils (Hansel/Wright stain) have POOR sensitivity AND specificity for AIN (also positive in pyelonephritis, atheroembolic disease, prostatitis, cystitis) and are NO LONGER recommended as a diagnostic test. They neither confirm nor exclude AIN. Renal biopsy remains the definitive test when the diagnosis is uncertain.

BOARD TRAP

Ordering urinary eosinophils to confirm or rule out AIN is outdated — a negative result does not exclude it and a positive result is nonspecific; relying on the test is the trap.

9. Sterile pyuria + WBC casts

WHAT YOU SEE

WBC casts drawn in flask

WHAT IT ENCODES

The characteristic urinalysis in AIN is STERILE PYURIA (white cells with negative culture), WBC CASTS, hematuria, and mild (sub-nephrotic) proteinuria. WBC casts point to tubulointerstitial inflammation. Eosinophil casts are NOT a reliable finding. Contrast this with the muddy-brown granular casts of ATN and the RBC/dysmorphic-RBC casts of glomerulonephritis.

BOARD TRAP

Sterile pyuria + WBC casts in the right drug context is AIN — misreading it as a 'culture-negative UTI/pyelonephritis' and treating with more antibiotics (potentially the offending drug class) is the trap.

10. 7-10 days latency

WHAT YOU SEE

Hourglass labeled 7-10 DAYS

WHAT IT ENCODES

Typical latency from drug exposure to AIN is days to weeks — classically ~7–10 days after FIRST exposure, but it can be as short as 1–2 days on RE-exposure (sensitized) and as long as weeks-to-months for NSAIDs/PPIs. The temporal link to a new drug is a key diagnostic clue; always build a careful medication timeline.

BOARD TRAP

AKI appearing only a couple of days after starting a drug doesn't argue against AIN if the patient was previously sensitized — accelerated onset on re-exposure is expected, so a short latency does not exclude AIN.

11. Stop drug — first step

WHAT YOU SEE

Ladder: STOP DRUG / OBSERVE 1 WEEK

WHAT IT ENCODES

The single most important first-line management step is to IDENTIFY and STOP the offending drug — many patients recover renal function spontaneously after withdrawal. Observe for improvement over roughly a week. Avoid re-introducing the culprit (and cross-reacting agents). Provide supportive care for the AKI in the interim.

BOARD TRAP

Adding steroids while continuing the offending drug is wrong — discontinuation of the culprit is the essential first step; steroids without stopping the drug will not resolve AIN.

12. Steroids ± dialysis

WHAT YOU SEE

Ladder: STEROIDS / DIALYSIS / DELAY=WORSE

WHAT IT ENCODES

If renal function does not recover after stopping the drug (or if biopsy shows severe/extensive inflammation), give GLUCOCORTICOIDS (e.g., prednisone ~0.5–1 mg/kg/day, or IV methylprednisolone pulse for severe disease, tapered over several weeks). EARLY treatment matters — delayed steroid initiation (beyond ~1–2 weeks, or progression to fibrosis) predicts worse, sometimes irreversible, renal recovery. Dialysis is supportive for severe AKI/uremia.

BOARD TRAP

Indefinitely 'watching' a non-recovering AIN before starting steroids is harmful — delaying glucocorticoids allows interstitial fibrosis and permanent CKD; timely steroids are key once drug withdrawal alone fails.

13. Rifampin re-challenge

WHAT YOU SEE

Banner 'RIFAMPIN RE-CHALLENGE' with dragon

WHAT IT ENCODES

Rifampin is a special, severe cause of immune AIN, classically triggered by INTERMITTENT or interrupted-then-resumed dosing ('re-challenge'). It is associated with rifampin-dependent antibodies and can produce an abrupt, severe immunoallergic reaction (fever, AKI, sometimes hemolysis/thrombocytopenia/flu-like syndrome). Never re-challenge with rifampin after a suspected reaction.

BOARD TRAP

Restarting rifampin after a prior interruption (assuming the earlier reaction was unrelated) can precipitate fulminant immune AIN — re-challenge is contraindicated, and treating the reaction as a simple 'flu' is the trap.